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Federal Contract Flows in Maryland

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Executive Summary

This report examines the flow of federal contract dollars into and out of Maryland through subcontracting relationships during fiscal years 2010–2024. While prime contract awards are typically measured by the location of the winning firm’s headquarters, the actual economic impact of federal procurement depends critically on where subcontracted work is performed. Our analysis tracks two key measures: **subcontract inflow**: federal contracts that flow into Maryland when out-of-state prime contractors hire Maryland firms as subcontractors, and **subcontract outflow**: federal contracts that leave Maryland when in-state primes subcontract work to vendors outside the state.

Key Findings:

- **Maryland shifted from net recipient to net sender.** Maryland was a net recipient of subcontract dollars from 2014 to 2019, peaking at \$1 billion in 2016. By 2024, this reversed to a net outflow of \$1.2 billion.
- **Defense brings dollars in; energy sends them out.** The Department of Defense generates \$10 billion in net inflow for Maryland. The Department of Energy shows the opposite: a net outflow of \$8 billion.
- **Technical work flows in; support services flow out.** Maryland receives subcontracts in Aerospace (\$7B), Computer Systems Design (\$5B), and Scientific R&D (\$4B). The largest outflow is Facilities Support Services (\$9B).
- **Subcontracts concentration.** Northrop Grumman Systems Corp leads with \$230 million annually, over four times the second-place recipient.

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1 Introduction

Federal procurement often operates through a prime–subcontracting structure. A federal agency awards a prime contract to a principal contractor, who then executes the work by allocating portions of performance to downstream suppliers and service providers through subcontracts. In practice, subcontracting is a key channel through which federal dollars reach local firms, specialized vendors, and smaller businesses that do not directly hold prime awards. As a result, the geography of prime awards can differ materially from the geography of actual production and economic activity, because prime contractors may be headquartered in one place while subcontracts—and the associated employment and spending—are distributed elsewhere.

Our analysis dissects Maryland’s position in federal contract flows by tracking subcontract movements across state borders. We define two key measures:

- **Subcontract Outflow.** Dollars awarded through prime contracts to Maryland firms that subsequently subcontract work outside the state, representing federal spending that leaks from Maryland’s economy to out-of-state suppliers.
- **Subcontract Inflow.** Dollars from prime contracts awarded to non-Maryland firms that flow into Maryland through subcontracts to in-state vendors, reflecting the state’s role as a downstream participant in national procurement supply chains.

2 Aggregate Trends of Contract in Maryland

Figure 1 summarizes Maryland’s federal subcontract inflow and outflow from FY2010 to FY2024. Both series grow over the sample period, but their relative magnitudes shift considerably. Inflow rises steadily through the 2010s and peaks at roughly \$4 billion in FY2022, before declining to approximately \$2 billion by FY2024. Outflow, by contrast, remains below inflow for most of the decade but accelerates after 2020, reaching \$3 billion in FY2024.

The net flow panel captures this divergence more directly. Between FY2010 and 2013, the net subcontract flow was near zero as both series remained small. Maryland then emerges as a consistent net recipient from FY2014 through 2019, with net inflow peaking around \$1 billion in FY2016. This trend reverses sharply after 2020: net flow turns negative and widens to approximately $-\$1.2$ billion by FY2024.

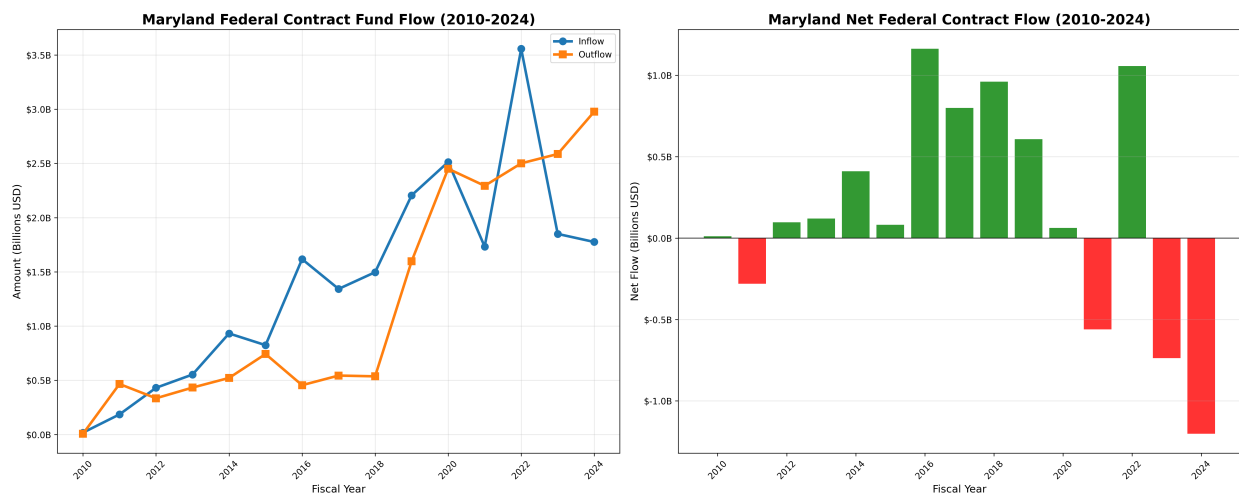


Fig. 1: Federal Contract Flows to Maryland Over Time

3 State-Level Decomposition

Figure 2 disaggregates cumulative subcontract flows by counterpart state for FY2024. On the inflow side, Virginia is the dominant source of subcontract dollars flowing into Maryland, totaling \$8 billion, followed by Texas (\$5 billion) and intra-state subcontracting within Maryland itself (\$4 billion). California, Ohio, and a cluster of states, including New Jersey, Missouri, Alabama, Tennessee, and Florida, each contribute between \$0.5 billion and \$1 billion.

On the outflow side, the pattern is more concentrated. Tennessee is the largest recipient of subcontract dollars, absorbing \$7 billion in FY2024. Intra-state flows and Virginia follow closely at \$4 billion, respectively. Texas (\$2 billion) and California (\$2 billion) round out the top destinations, while Pennsylvania, the District of Columbia, New York, Florida, and Massachusetts each receive between \$0.5 billion and \$1 billion.

The Virginia–Maryland corridor dominates both directions, reflecting the tight integration of the two states within the Washington, D.C. metropolitan procurement ecosystem.

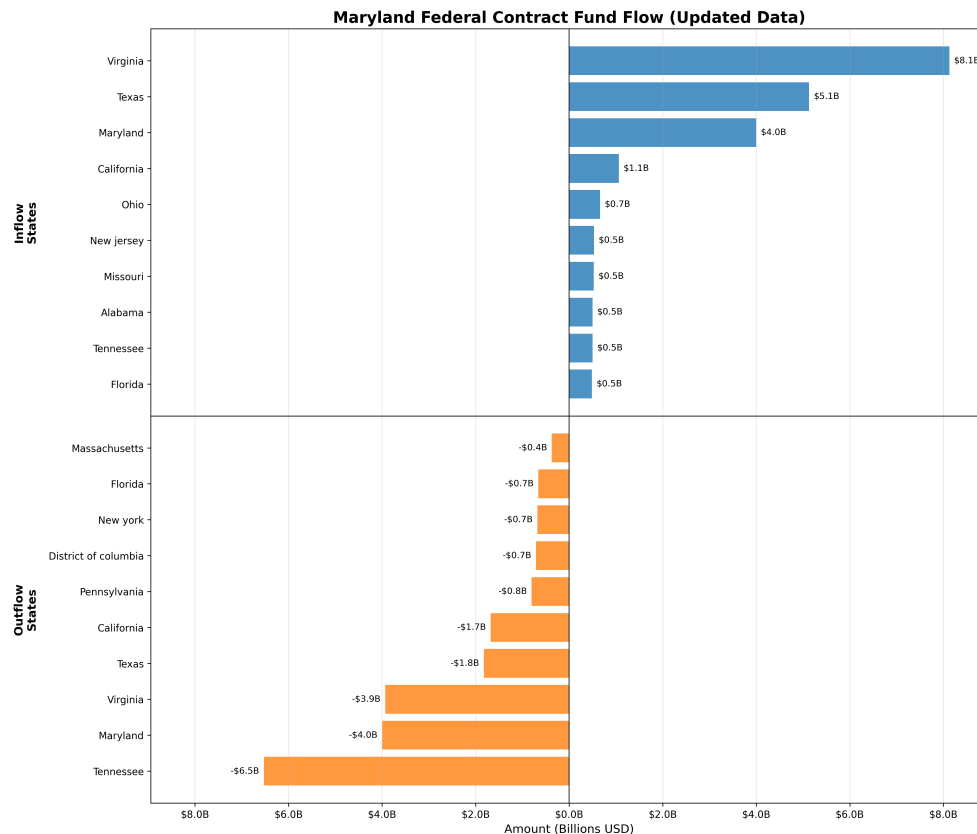


Fig. 2: County Distribution of Federal Contract Flows to Maryland

4 Agency-Level Decomposition

Figure 3 ranks the top five federal agencies by cumulative subcontract inflow and outflow associated with Maryland at FY2024. The Department of Defense (DOD) dominates both sides of the ledger, generating \$19 billion in subcontract inflow to Maryland against \$9 billion in outflow, yielding a net inflow of approximately \$10 billion.

Beyond DOD, the remaining agencies are considerably smaller in scale but reveal important asymmetries. The Department of Health and Human Services (HHS) and NASA each bring roughly \$2 billion into Maryland through subcontracts, while their outflows reach \$2 billion and \$3 billion respectively, making both approximately net neutral for the state. The Department of Energy (DOE) presents the starkest imbalance: inflow amounts to under \$1 billion against \$8 billion in outflow, producing a net loss of over \$7 billion. The Department of Homeland Security (DHS) is relatively balanced, with \$1 billion in inflow and \$1 billion in outflow.

Taken together, Maryland's favorable aggregate position in defense subcontracting masks substantial leakage through energy-related contracts.

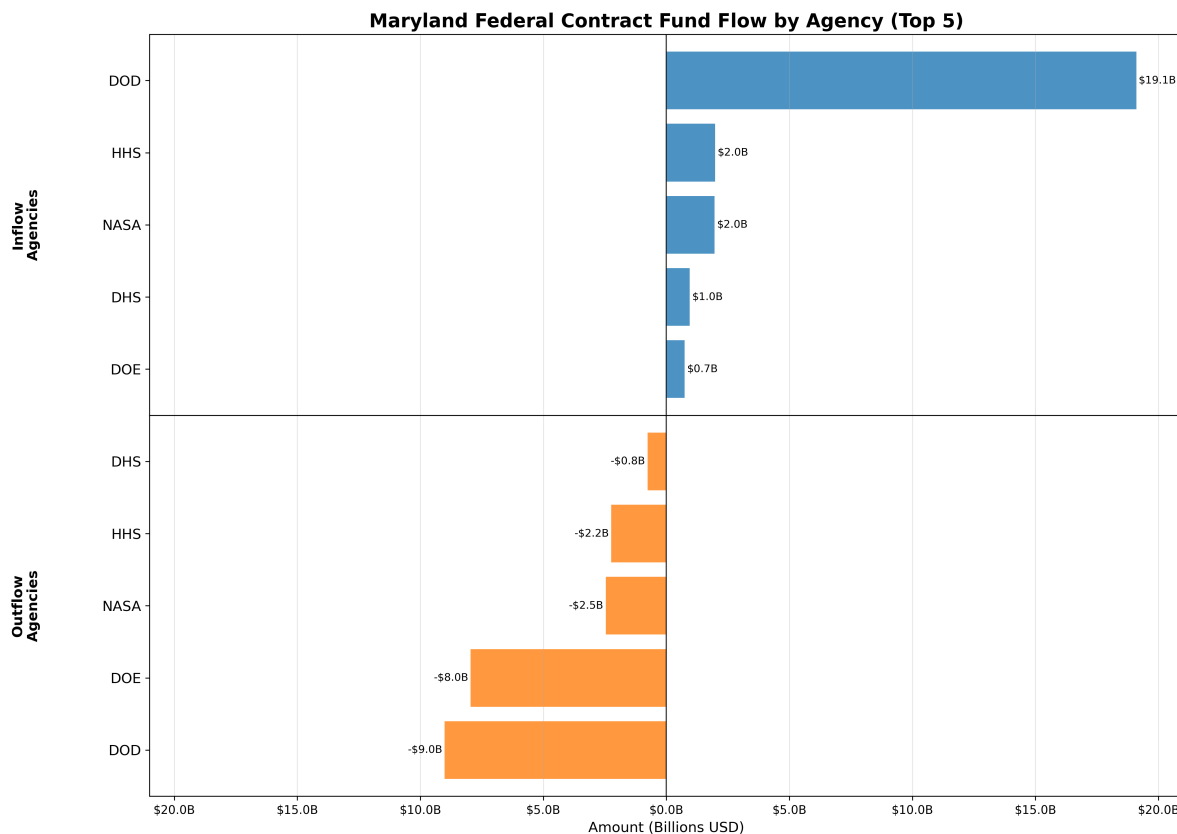


Fig. 3: Federal Contract Flows to Maryland by Awarding Agency

5 Industry-Level Decomposition

Figure 4 presents subcontract flows by industry sector using 4-digit NAICS codes. First, inflows concentrate in high-value technical sectors, with Aerospace Product and Parts Manufacturing (\$7B), Computer Systems Design and Related Services (\$5B), and Scientific Research and Development Services (\$4B) representing the largest inflow categories, indicating that Maryland's high-tech industries receive substantial subcontract work from out-of-state prime contractors. Second, outflows dominate in facilities support, with Facilities Support Services (\$9B) constituting the largest outflow category, suggesting that Maryland primes rely heavily on out-of-state vendors for building operations and maintenance services.

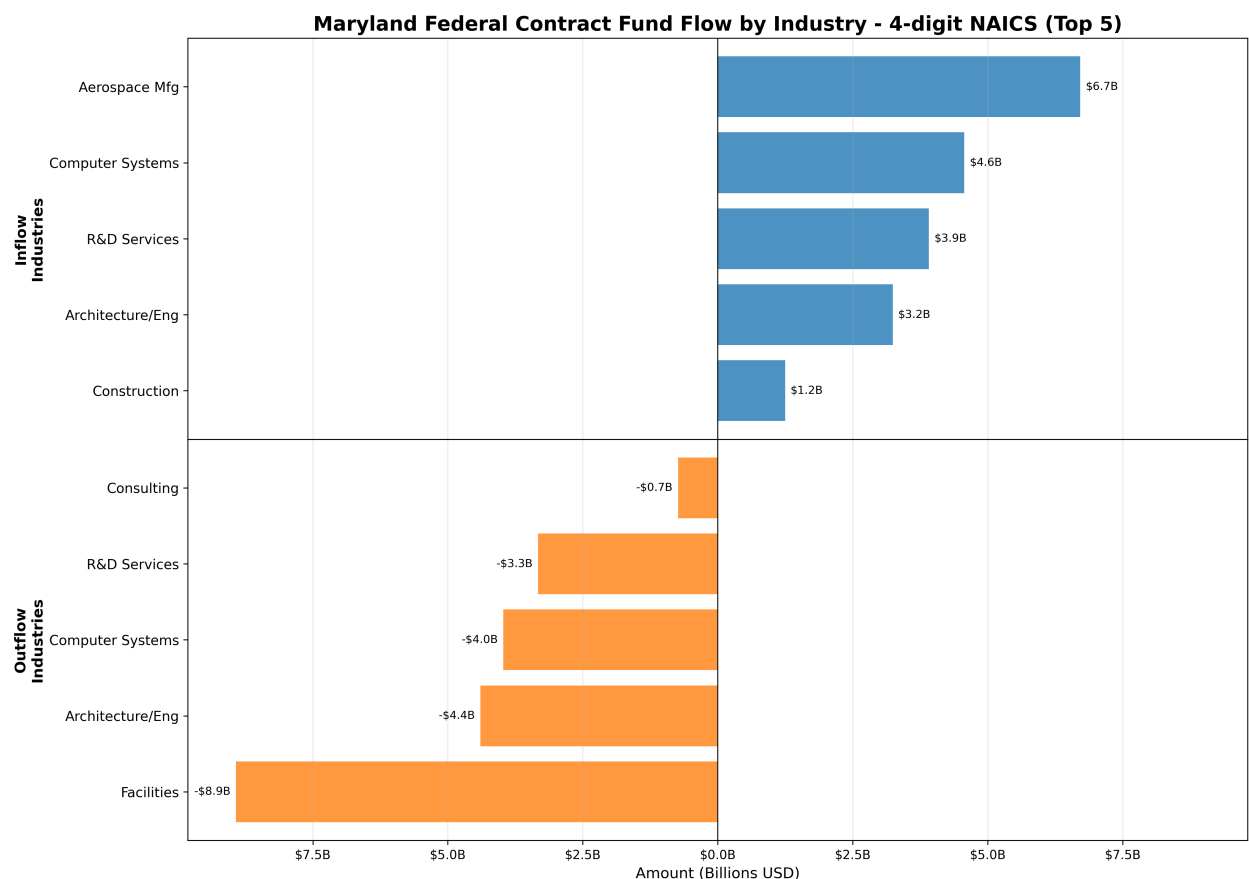


Fig. 4: Federal Contract Flows to Maryland by Industry (4-Digit NAICS)

6 Contractor-Level Decomposition

Figures 5 and 6 present the top 10 federal subcontract recipients located in Maryland based on the average value from 2010–2024.

6.1 Top Subawardees by Annual Average

Figure 5 ranks Maryland subawardees by annual average subcontract amount. Northrop Grumman Systems Corp dominates the rankings with an annual average of \$230 million, far exceeding the second-place TEKsystems (\$50 million) and third-place Textron Systems Corporation (\$50 million). Other major defense contractors follow: Northrop Grumman Corporation (\$36 million), Lockheed Martin Corp (\$35 million), and FCN, Inc. (\$34 million). The remaining top 10 include Edgewater Federal Solutions (\$29 million), Smiths Detection Inc. (\$23 million), Motorola Solutions (\$22 million), and IBM (\$21 million). This concentration underscores that federal subcontracting dollars flow disproportionately to a small number of large contractors in the state.

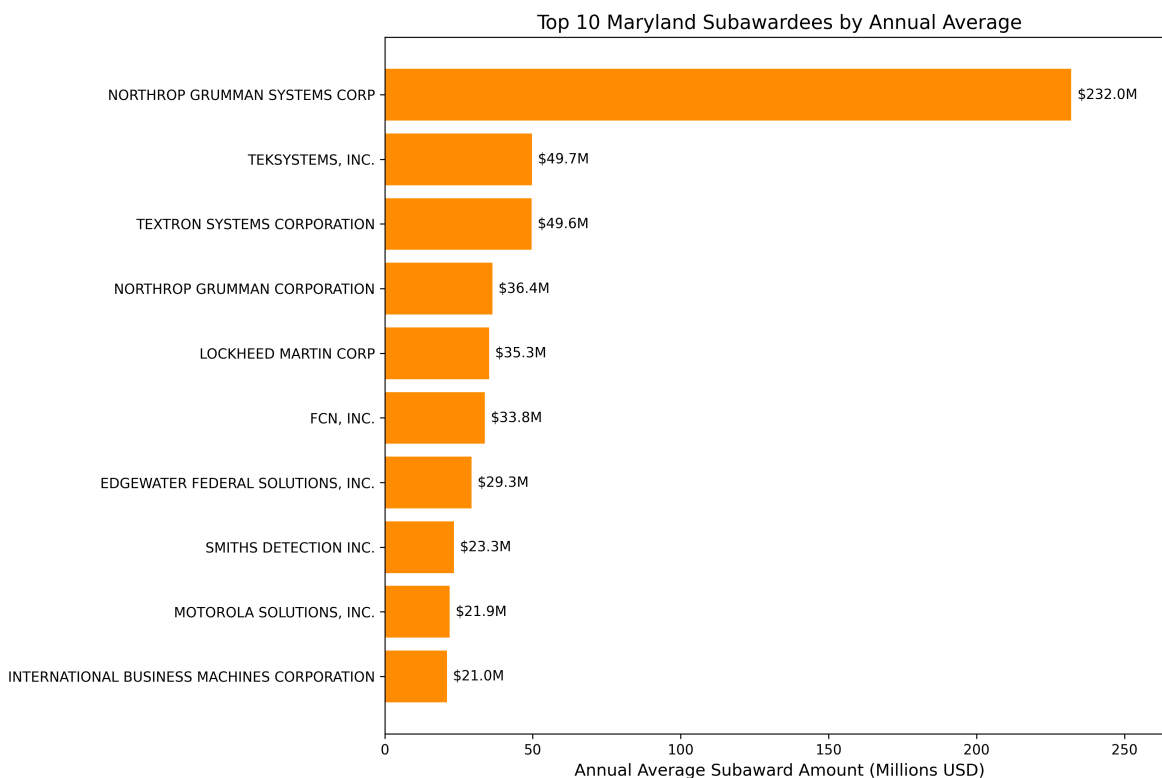


Fig. 5: Top 10 Maryland Subawardees by Annual Average Amount.

6.2 Funding Sources by Agency

Figure 6 illustrates the distribution of federal contracts for the top 10 Maryland sub-awardees. The Department of Defense (DoD) is the dominant funding source, accounting for the vast majority of subcontract awards to Northrop Grumman Systems Corp (\$3.5 billion total), TEKsystems, Textron Systems, Northrop Grumman Corporation, FCN, and Motorola Solutions. The Department of Homeland Security (DHS) serves as the primary funding source for Lockheed Martin Corp (\$530 million) and Smiths Detection Inc. (\$350 million), reflecting their focus on security-related contracts. Notably, Edgewater Federal Solutions (\$440 million) receives nearly all of its funding from the Department of Energy, while IBM (\$320 million) exhibits a more diversified funding portfolio spanning the Department of Veterans Affairs, DoD, and Health and Human Services.

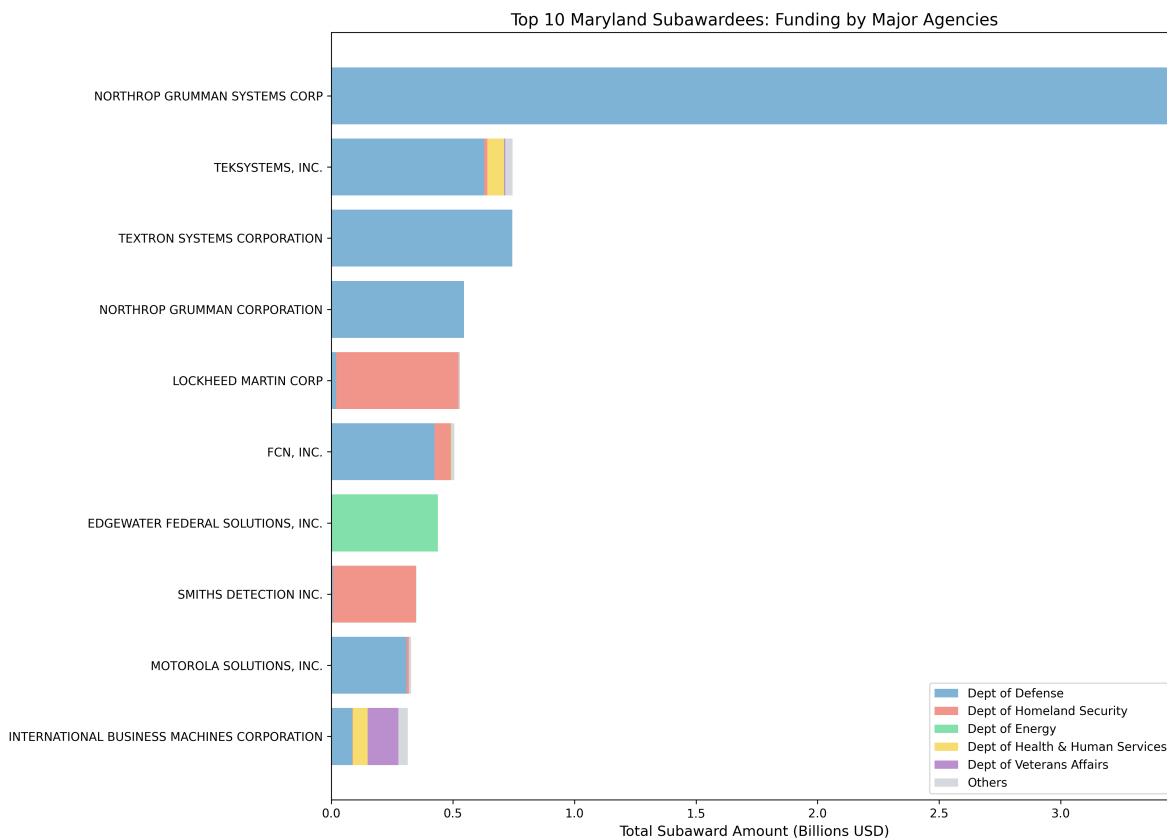


Fig. 6: Top 10 Maryland Subawardees: Funding by Major Agencies.

7 County-Level Decomposition

Figure 7 presents average annual subcontract inflow, outflow, and net flow by Maryland county over FY2015–2024. Anne Arundel County leads all counties in inflow, averaging over \$700 million per year, followed by Montgomery County at roughly \$400 million and Howard County at approximately \$160 million. Prince George’s County, Baltimore County, and a group of smaller counties, including Harford, Frederick, St. Mary’s, Carroll, and Baltimore City, each average between \$50 million and \$110 million annually.

The outflow distribution is notably more concentrated. Frederick County dominates with an average annual outflow approaching \$750 million, far exceeding all other counties. Montgomery County follows at roughly \$450 million, with Howard and Prince George’s counties each near \$100 million. The remaining counties on the outflow side, including Baltimore, St. Mary’s, Anne Arundel, Harford, Baltimore City, and Cecil, contribute relatively modest amounts.

The net flow panel clarifies which counties serve as net recipients versus net senders of subcontract dollars. Anne Arundel County stands out as the largest net beneficiary, with an average annual net inflow exceeding \$600 million, reflecting its role as a major downstream destination for subcontracted federal work. On the other side, Frederick County is the largest net sender, with average annual net outflow surpassing \$600 million, followed by Montgomery County at roughly \$50 million.

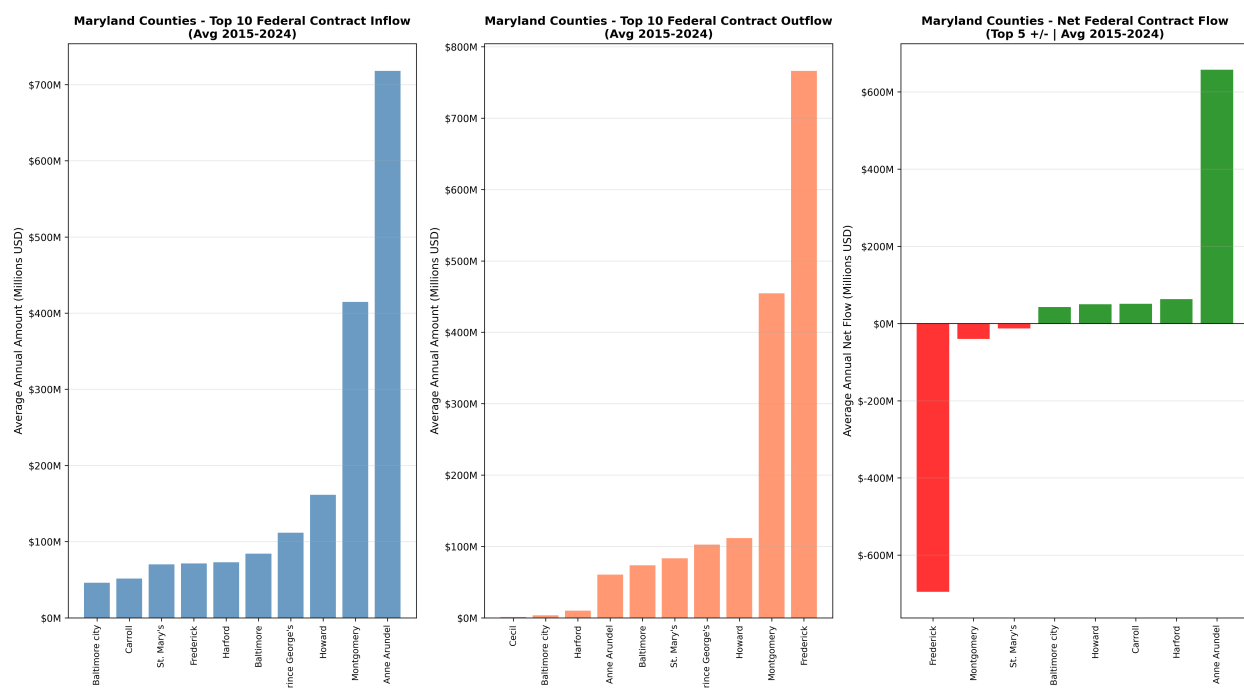


Fig. 7: Top 10 Maryland Counties by Federal Contract Flows

8 Top-3 County Decomposition

To better understand the micro-geography of Maryland's subcontract flows, we examine three of the state's most active counties in federal procurement: Anne Arundel, Howard, and Montgomery. Each exhibits a distinct pattern in terms of geographic counterparts, agency composition, and net position.

8.1 Anne Arundel County

Anne Arundel County is the largest net recipient of subcontract dollars among Maryland counties, and its flow profile is heavily concentrated along a single axis. On the inflow side, Texas alone accounts for \$0.4 billion in average annual subcontracts directed to Anne Arundel vendors, dwarfing the next largest source, Virginia, at \$0.2 billion. Intra-state flows from other Maryland counties and smaller contributions from Washington and Tennessee round out the top five. By agency, the DOD overwhelmingly dominates, generating \$0.7 billion in average annual inflow, more than thirty times the next largest agency (DOE at \$0.02 billion). Outflows from Anne Arundel are minimal, with the largest destination being intra-state transfers at just \$0.02 billion. The county's strong net positive position stems almost entirely from downstream DOD subcontracts awarded by out-of-state primes, particularly from Texas.

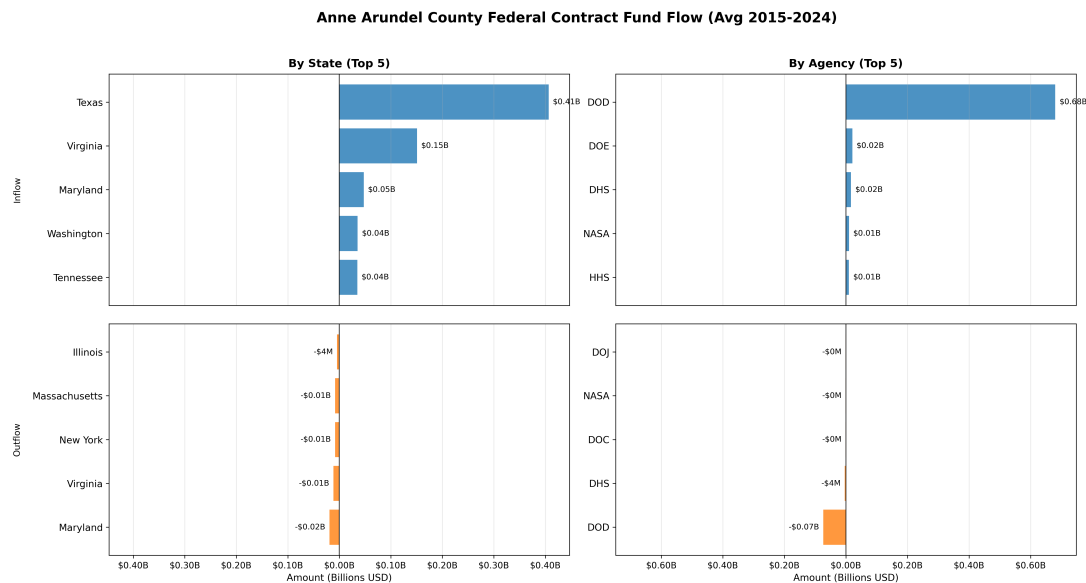


Fig. 8: Federal Contract Flows to Anne Arundel County

8.2 Howard County

Howard County presents a more balanced but still net-positive profile. Virginia is the leading source of inflow at \$0.08 billion annually, followed by intra-state flows (\$0.04 billion) and smaller contributions from Massachusetts, Texas, and Ohio. DOD again leads on the agency side with \$0.12 billion in inflow, supplemented by HHS and NASA at \$0.02 billion each. On the outflow side, however, Howard County sends a notable volume of subcontract dollars to other Maryland counties (\$0.04 billion) and to Virginia (\$0.02 billion). By agency, NASA is the largest source of outflow at \$0.07 billion, followed by DOD at \$0.04 billion and HHS at \$0.03 billion. This suggests that its contractors function as intermediaries, receiving DOD subcontracts while simultaneously farming out NASA and HHS work to vendors elsewhere.

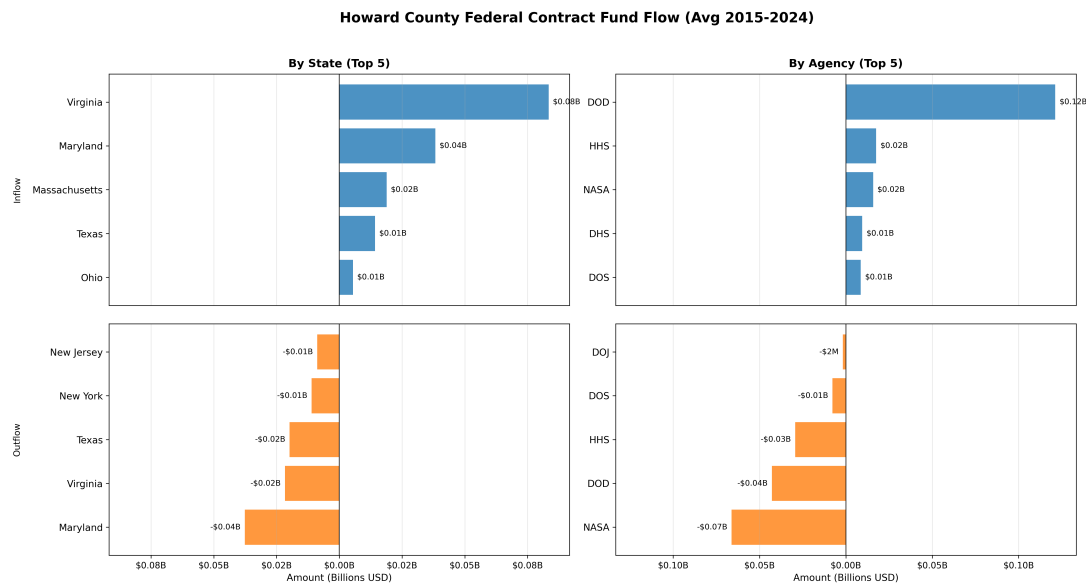


Fig. 9: Federal Contract Flows to Anne Howard County

8.3 Montgomery County

Montgomery County is the most active county in terms of bilateral flow volume, but its net position is approximately neutral. Virginia is the dominant inflow source at \$0.3 billion, with intra-state, Ohio, Mississippi, and Texas contributing between \$0.03 billion and \$0.07 billion each. DOD leads inflow by agency at \$0.23 billion, followed by DHS and HHS at \$0.06 billion apiece. Outflows, however, are substantial: Virginia receives \$0.22 billion and intra-state transfers account for \$0.13 billion, with Massachusetts, Florida, and Texas absorbing smaller amounts. On the agency side, DOD outflow reaches \$0.34 billion, exceeding its inflow, while HHS (\$0.06 billion) and DHS (\$0.05 billion) also register meaningful outflows.

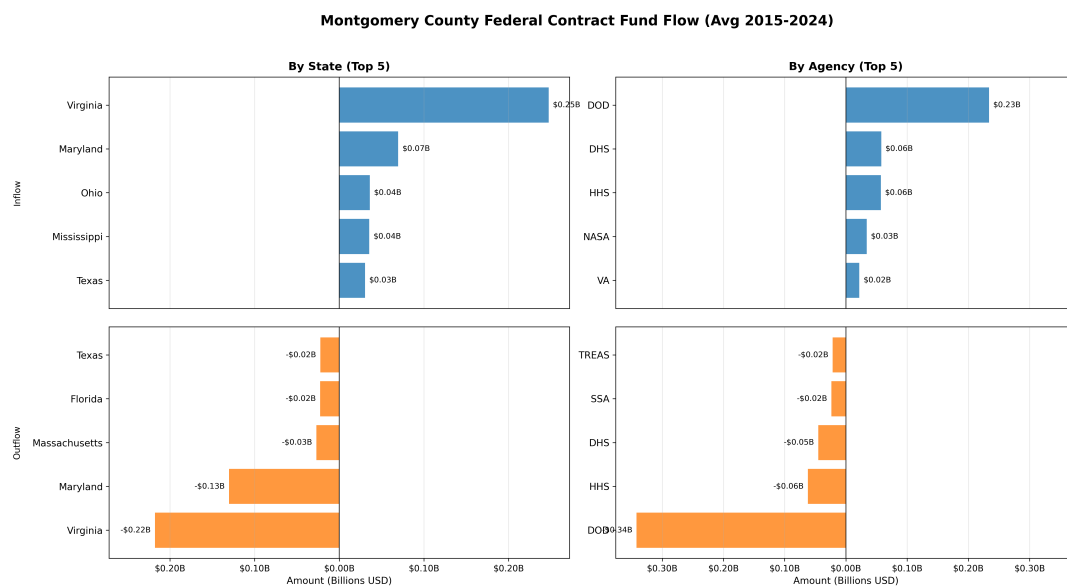


Fig. 10: Federal Contract Flows to Montgomery County